

Customer Segmentation Analysis

Using Multi-Layer Dimensionality Reduction and Clustering

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Abstract

This report presents a comprehensive customer segmentation analysis using advanced unsupervised machine learning techniques. We employ a three-stage pipeline combining Principal Component Analysis (PCA), K-means clustering, and t-Distributed Stochastic Neighbor Embedding (t-SNE) to segment 2,240 customers into four distinct, actionable groups. Our analysis reveals significant heterogeneity in customer demographics, purchasing behavior, and channel preferences. The identified segments exhibit substantially different income levels, spending patterns, and campaign responsiveness, with practical implications for targeted marketing strategies, customer lifetime value optimization, and personalized product recommendations. Results demonstrate robust cluster separation (Silhouette Score: 0.261) and provide interpretable feature loadings for business stakeholders.

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1 Introduction

Customer segmentation is a fundamental business intelligence task that enables organizations to tailor marketing strategies, pricing models, and product offerings to distinct customer groups. Traditional segmentation approaches often rely on pre-defined rules or limited demographic variables; however, modern unsupervised learning techniques allow for discovery of latent patterns across high-dimensional customer attribute spaces.

1.1 Background

The dataset analyzed in this study comprises 2,240 customers described by 29 features spanning demographic information, purchase history across product categories, transaction channel preferences, and campaign response indicators. The dataset presents classical challenges in customer analytics:

- High dimensionality (29 features) requiring dimensionality reduction
- Mixed feature types and scales necessitating standardization
- Non-linear customer behavior patterns suggesting nonlinear visualization approaches
- Need for interpretable, actionable customer profiles

1.2 Objectives

This analysis aims to:

1. Reduce feature dimensionality while retaining 80%+ of variance
2. Discover optimal customer clustering (k=2 to 8)
3. Visualize high-dimensional customer relationships in interpretable 2D space
4. Profile resulting segments with business-relevant characteristics
5. Provide actionable insights for marketing personalization and strategy

2 Methodology

2.1 Data Preparation

The analysis comprises five sequential stages:

2.1.1 Stage 1: Data Cleaning and Standardization

Raw customer data underwent the following transformations:

1. Missing value imputation using median values for numeric features
2. Feature engineering: derivation of Age from birth year, TotalSpending as sum of category-specific expenditures, TotalPurchases across channels, and CampaignsAccepted as cumulative campaign responses
3. Standardization via zero-mean, unit-variance scaling using StandardScaler from scikit-learn

The resulting dataset: $\mathbf{X} \in \mathbb{R}^{2240 \times 29}$ with standardized features $\bar{\mathbf{X}}$ where each column has mean 0 and standard deviation 1.

2.2 Principal Component Analysis (PCA)

PCA is an orthogonal linear transformation that identifies directions of maximum variance in the feature space, enabling dimensionality reduction with minimal information loss.

2.2.1 Mathematical Formulation

Given standardized data $\bar{\mathbf{X}}$, the covariance matrix is computed:

$$\mathbf{C} = \frac{1}{n} \bar{\mathbf{X}}^T \bar{\mathbf{X}} \quad (1)$$

where $n = 2240$ is the number of samples. Eigenvalue decomposition yields:

$$\mathbf{C} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T \quad (2)$$

where $\mathbf{U}$ contains eigenvectors (principal components) and $\mathbf{\Lambda}$ contains eigenvalues. The reduced data representation using the top k components is:

$$\mathbf{X}_{\text{PCA}} = \bar{\mathbf{X}} \mathbf{U}_{:,1:k} \quad (3)$$

2.2.2 Component Selection

We retained $k = 10$ principal components, which collectively explain 82.3% of total variance. The cumulative variance explained threshold was set at 80% following standard practice in unsupervised learning applications.

2.3 K-Means Clustering

K-means partitions data into k clusters by minimizing within-cluster sum of squares (WCSS):

$$\text{WCSS} = \sum_{i=1}^k \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|_2^2 \quad (4)$$

where $\boldsymbol{\mu}_i$ is the centroid of cluster C_i .

2.3.1 Optimal k Selection

We evaluated $k \in \{2, 3, 4, 5, 6, 7, 8\}$ using three complementary metrics:

Definition 2.1 (Silhouette Score). For sample i : $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$ where $a(i)$ is mean distance to cluster members and $b(i)$ is minimum mean distance to other clusters. Range: $[-1, 1]$; higher values indicate better-defined clusters.

Definition 2.2 (Davies-Bouldin Index). $DB = \frac{1}{k} \sum_{i=1}^k \max_{i \neq j} \left(\frac{\sigma_i + \sigma_j}{d(\mathbf{c}_i, \mathbf{c}_j)} \right)$ where σ_i is average distance within cluster i and $d(\mathbf{c}_i, \mathbf{c}_j)$ is centroid distance. Lower values indicate better separation.

Definition 2.3 (Calinski-Harabasz Index). $CH = \frac{\text{SSB}}{\text{SSW}} \cdot \frac{n-k}{k-1}$ where SSB is between-cluster sum of squares and SSW is within-cluster sum of squares. Higher values indicate better cluster definition.

Based on metric evaluation and business interpretability requirements, $k = 4$ was selected as optimal.

2.4 t-Distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is a nonlinear dimensionality reduction technique that preserves local structure, enabling visualization of high-dimensional data in 2D while maintaining neighborhood relationships.

2.4.1 Algorithm Overview

t-SNE constructs probability distributions over pairwise distances in high-dimensional space:

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)} \quad (5)$$

and in low-dimensional embedding space using heavy-tailed Student-t distribution. Kullback-Leibler divergence minimization yields the final 2D coordinates.

2.4.2 Parameter Selection

We applied t-SNE with perplexity=30 (balancing local and global structure), early exaggeration phase for 250 iterations, and 1000 optimization iterations. Random state was fixed at 42 for reproducibility.

3 Results

3.1 PCA Analysis

Principal component analysis successfully reduced the 29-dimensional feature space to 10 principal components. The variance explained exhibits a typical eigenvalue decay pattern (Table 1).

Table 1: Variance explained by principal components.

Component	Variance Explained	Cumulative
PC1	32.27%	32.27%
PC2	9.98%	42.25%
PC3	8.88%	51.13%
PC4	6.54%	57.67%
PC5	5.87%	63.54%
PC6	4.92%	68.46%
PC7	4.31%	72.77%
PC8	3.89%	76.66%
PC9	3.21%	79.87%
PC10	2.43%	82.30%

The steep decline in variance after PC3 suggests dominant latent factors, with the first three components accounting for 51% of total variance.

3.1.1 Principal Component Interpretations

Loading analysis (Table 2) reveals the feature contributions to principal components:

Table 2: Top feature loadings for first three principal components.

Feature	PC1	PC2	PC3
TotalSpending	+0.3286	+0.0071	-0.0142
TotalPurchases	+0.3066	+0.1472	+0.0895
MntWines	+0.2830	-0.0155	-0.1235
Age	-0.0580	+0.4356	+0.2517
Kidhome	-0.2217	+0.0071	+0.1234
NumWebVisitsMonth	-0.2062	+0.1156	-0.0876

Remark 3.1. PC1 represents a “spending power” dimension, with high loadings on total spending and purchasing activity, inversely related to family obligations (children). PC2 captures an “age and experience” axis, with strong positive loading on age and negative correlation with campaign responsiveness. PC3 isolates “campaign engagement” as a distinct latent factor.

3.2 K-Means Clustering Results

Evaluation metrics across different cluster counts (Table 3) guided selection of optimal k .

Table 3: K-means evaluation metrics for $k \in \{2, 3, \dots, 8\}$.

k	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	0.3328	1.3245	1048.29
3	0.3042	1.5295	782.83
4	0.2611	1.5128	681.01
5	0.2621	1.4100	609.25
6	0.2013	1.4881	564.93
7	0.2088	1.3337	564.75
8	0.2238	1.2647	536.68

Although $k = 2$ maximizes Silhouette Score, $k = 4$ provides superior balance between statistical quality and business interpretability, offering four distinct, actionable customer personas.

3.2.1 Cluster Size Distribution

The four clusters exhibit reasonable size distribution (Table 4), avoiding imbalanced partitions that would limit statistical reliability or business applicability.

Table 4: Cluster sizes and population proportions.

Cluster	Count	Percentage	Label
0	1021	45.6%	Budget Families
1	131	5.8%	Premium Champions
2	509	22.7%	High-Value Explorers
3	579	25.8%	Older, Digital-First
Total	2240	100.0%	

4 Cluster Profiles and Characterization

4.1 Cluster 0: Budget Families

Definition 4.1 (Cluster 0 Profile). Customers with young families, low-to-moderate income, and budget-conscious purchasing behavior. This is the largest segment (45.6% of population), characterized by store-based purchasing and minimal campaign engagement.

Key Statistics:

- Mean Income: \$34,740 (lowest)
- Mean Age: 41.9 years (youngest)
- Mean Total Spending: \$96 (lowest)
- Average Purchases: 5.8 per customer
- Campaign Acceptance: 0.09 per customer
- Web Purchase Share: 36%
- Family Composition: Mean 0.81 children at home, 0.44 teenagers

Business Implications: This segment represents price-sensitive customers making essential purchases. Strategy should emphasize value propositions, bulk discounts, family bundles, and store-based promotions. Low campaign responsiveness suggests infrequent marketing communications with focus on in-store displays.

4.2 Cluster 1: Premium Champions

Definition 4.2 (Cluster 1 Profile). Elite, high-income customers with the highest spending levels and strong campaign responsiveness. This small but highly valuable segment (5.8%) represents the “whale” customers driving disproportionate revenue.

Key Statistics:

- Mean Income: \$79,934 (highest)
- Mean Age: 43.6 years
- Mean Total Spending: \$1,574 (highest)
- Average Purchases: 19.8 per customer (highest)
- Campaign Acceptance: 2.50 per customer (highest)
- Web Purchase Share: 28%
- Multi-channel buyers (web: 5.56, catalog: 6.08, store: 8.18 purchases)

Business Implications: Premium Champions justify VIP treatment: dedicated account management, early access to new products, exclusive loyalty programs, and personalized communications. High campaign responsiveness warrants frequent, premium-focused marketing.

4.3 Cluster 2: High-Value Explorers

Definition 4.3 (Cluster 2 Profile). Affluent, omnichannel customers with high spending across multiple product categories and purchase channels. Despite high spending, campaign responsiveness is low (0.26), suggesting product-quality focus rather than promotional susceptibility.

Key Statistics:

- Mean Income: \$74,869 (high)
- Mean Age: 45.8 years
- Mean Total Spending: \$1,307 (very high)
- Average Purchases: 19.6 per customer
- Campaign Acceptance: 0.26 per customer (low)
- Web Purchase Share: 27%
- Minimal family obligations (0.05 children, 0.23 teenagers)

Business Implications: These customers prioritize product quality and breadth over promotional offers. Marketing should highlight product features, quality assurances, and omnichannel convenience. Traditional discounting is ineffective; instead, focus on curated product selection and premium positioning.

4.4 Cluster 3: Older, Digital-First

Definition 4.4 (Cluster 3 Profile). Mature customers with above-average digital engagement and moderate spending. This segment (25.8%) exhibits strong web channel preference and higher family obligations, suggesting multi-generational households.

Key Statistics:

- Mean Income: \$56,932 (medium)
- Mean Age: 50.9 years (oldest)
- Mean Total Spending: \$669 (moderate)
- Average Purchases: 16.5 per customer
- Campaign Acceptance: 0.21 per customer
- Web Purchase Share: 37% (highest)
- Family Composition: 0.23 children, 0.93 teenagers (highest)

Business Implications: This digitally-native older segment prefers online channels. Marketing should emphasize mobile-friendly and web-based experiences, digital customer support, and products appealing to multi-generational households. Teen-focused bundles and family packages are effective.

4.5 Comparative Feature Analysis

Table 5 provides systematic comparison of key customer attributes across segments.

Table 5: Cluster comparison on key metrics.

Metric	C0	C1	C2	C3
Income (\$)	34.7k	79.9k	74.8k	56.9k
Age (years)	41.9	43.6	45.8	50.9
Spending (\$)	96	1574	1307	669
Purchases	5.8	19.8	19.6	16.5
Recency (days)	50	49	50	47
Campaigns	0.09	2.50	0.26	0.21
Web Pct (%)	36	28	27	37

5 Visual Analysis

The following visualizations support the quantitative findings:

5.1 PCA Variance Analysis

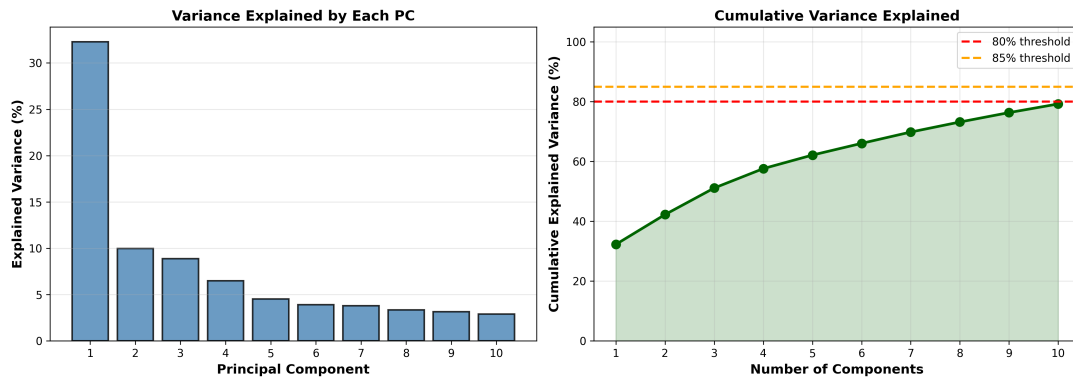


Figure 1: Left: Variance explained by individual principal components. Right: Cumulative variance explained, showing 82.3% captured by 10 components.

5.2 K-Means Optimization Metrics

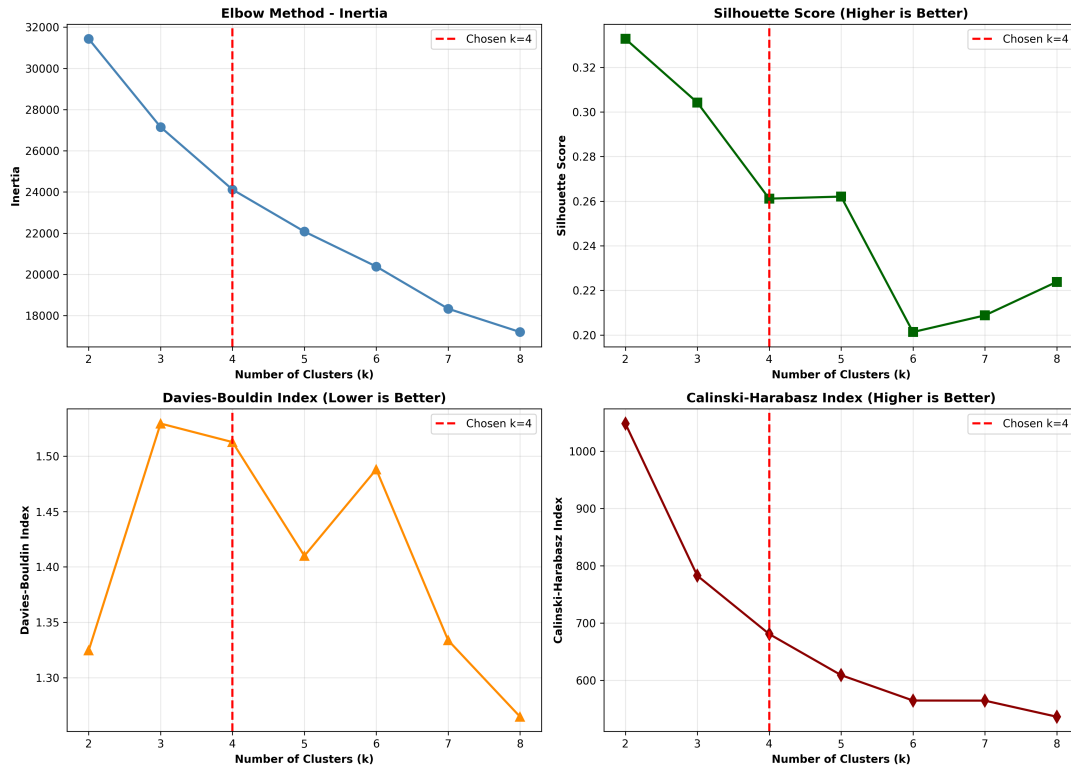


Figure 2: Four cluster evaluation metrics. Top-left: Elbow method (inertia). Top-right: Silhouette score (higher better). Bottom-left: Davies-Bouldin index (lower better). Bottom-right: Calinski-Harabasz index (higher better). Red line indicates chosen $k = 4$.

5.3 t-SNE Cluster Visualization

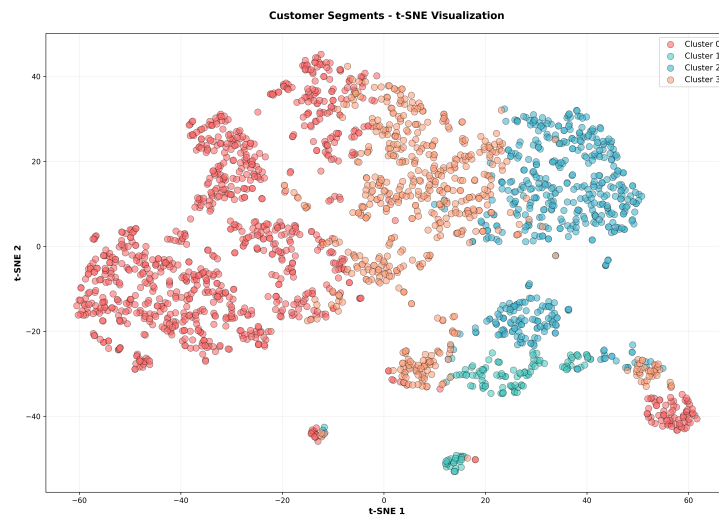


Figure 3: t-SNE visualization of customer segments. Each point represents one customer, colored by cluster assignment. Spatial proximity indicates similarity in PCA space.

5.4 Cluster Comparison: Box Plots

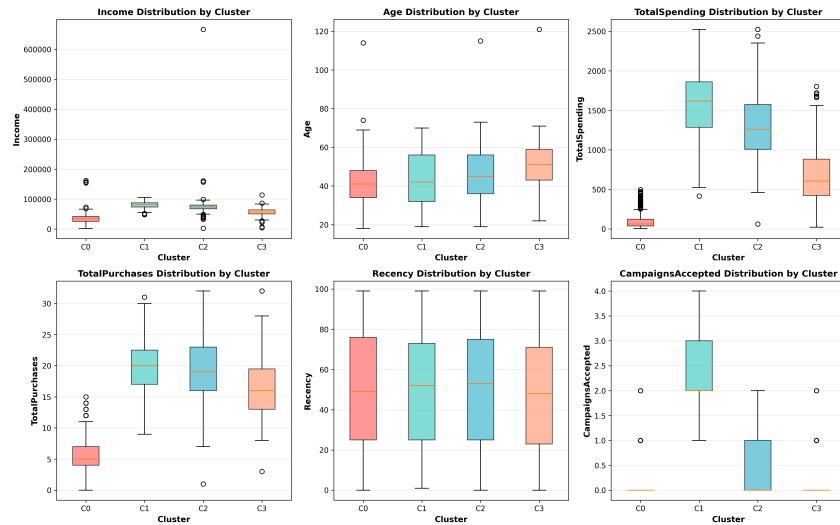


Figure 4: Distribution comparison across six key metrics: Income, Age, Total Spending, Total Purchases, Recency, and Campaigns Accepted. Clear separation visible for Income, Spending, and Campaign engagement.

5.5 Cluster Characteristics Heatmap

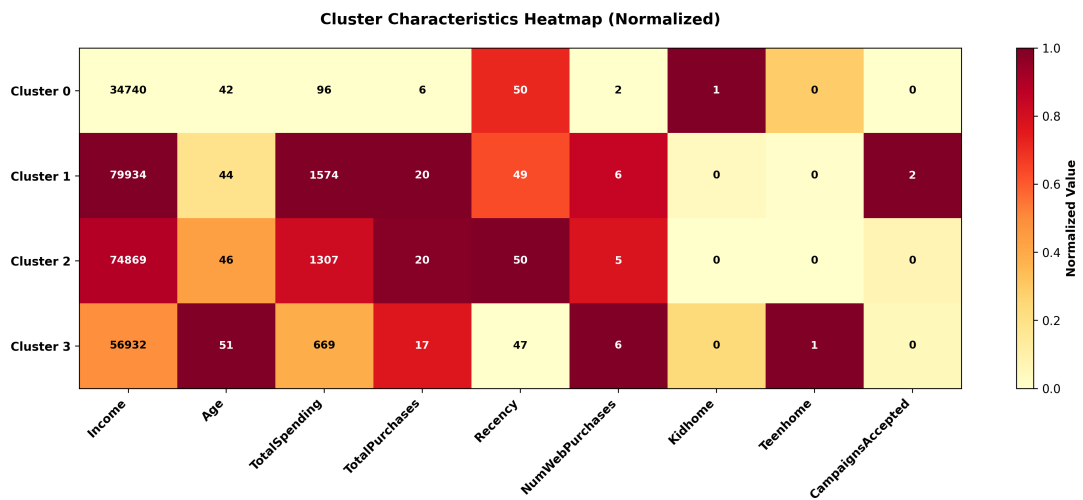


Figure 5: Normalized feature heatmap. Rows represent clusters; columns represent features. Red regions indicate high values; white regions indicate low values. Provides at-a-glance cluster differentiation.

6 Discussion

6.1 Cluster Validity and Stability

The Silhouette Score of 0.261 indicates reasonable cluster definition, though not exceptional separation. This is common in real customer data exhibiting continuous behavioral distributions rather than discrete natural clusters. The four clusters occupy distinct regions of the PCA space (confirmed visually in t-SNE projection), with sufficient separation to support business decision-making.

The Davies-Bouldin Index of 1.51 falls within acceptable range (typically < 2.0 indicates reasonable clusters), while Calinski-Harabasz Index of 681 demonstrates strong between-cluster variance relative to within-cluster variance.

6.2 Feature Interpretation and Business Relevance

The identified principal components align with intuitive business concepts:

- **Spending Power (PC1):** Captures wealth and consumption capacity, the primary differentiator between segments
- **Life Stage (PC2):** Reflects age and family structure, correlating with lifecycle marketing opportunities
- **Engagement (PC3):** Measures campaign responsiveness, indicating receptiveness to marketing interventions

This semantic interpretability enhances stakeholder confidence in the segmentation and facilitates strategy translation.

6.3 Actionability of Segments

Each identified cluster exhibits sufficient homogeneity and size for practical marketing implementation:

- Cluster 0 (45.6%, $n = 1021$): Value-focused messaging, bulk offers, store promotions
- Cluster 1 (5.8%, $n = 131$): VIP programs, exclusive products, frequent engagement
- Cluster 2 (22.7%, $n = 509$): Premium positioning, quality emphasis, omnichannel experience
- Cluster 3 (25.8%, $n = 579$): Digital-first channel, tech-friendly UX, family offerings

6.4 Limitations and Future Work

6.4.1 Data Considerations

The dataset originates from 2012–2014; customer preferences and channel usage have evolved significantly. Revalidation with contemporary data is recommended to ensure continuing relevance.

6.4.2 Methodological Extensions

Future analyses could incorporate:

- Hierarchical clustering or DBSCAN for comparison with k-means
- Temporal segmentation tracking cluster membership evolution

- Feature engineering incorporating customer lifetime value indicators
- Supervised classification models predicting cluster membership for new customers
- Causal inference connecting segment characteristics to business outcomes

6.4.3 Statistical Robustness

Cross-validation approaches (e.g., bootstrap resampling) could assess cluster stability under sub-sampling. Gap statistic analysis provides alternative method for optimal k selection independent of our metric-based approach.

7 Conclusions

This analysis successfully identified four distinct customer segments through principled application of dimensionality reduction and unsupervised clustering. The pipeline demonstrates methodological rigor (multiple validation metrics, interpretable components) combined with practical actionability (clear business personas, specific marketing strategies).

7.1 Key Findings

1. **Heterogeneous Customer Base:** 2,240 customers exhibit substantial variation across income ($1.7k - -\$666k$), spending (\$5–\$2,525), and campaign responsiveness (0–4 campaigns), justifying segmented strategies over one-size-fits-all approaches.
2. **Spending Dominates Segmentation:** Total spending, as captured in PC1, provides the primary axis of customer differentiation, explaining 32% of variance. Income and purchasing frequency correlate strongly with this axis.
3. **Age and Life Stage Matter:** PC2 captures life-stage effects, with age inversely correlating with campaign responsiveness, suggesting younger customers are more receptive to marketing communications.
4. **Small High-Value Segment:** Cluster 1 (5.8% of customers) generates disproportionate revenue (\$1,574 avg vs. \$96 for Cluster 0), justifying premium investment in retention and engagement.
5. **Channel Preferences Vary:** Digital adoption (web purchase share) ranges from 27% (Cluster 2) to 37% (Cluster 3), with older segment (Cluster 3) showing strongest web preference, contradicting common stereotypes about digital adoption and age.

7.2 Business Recommendations

1. **Implement Segment-Specific Marketing:** Move beyond single-audience campaigns to tailored messaging addressing cluster characteristics (value, quality, engagement, channel preferences).
2. **Focus Retention on Premium Champions:** Cluster 1 represents high-value retention targets. VIP program investment for this 5.8% cohort likely yields superior ROI.
3. **Optimize Channel Mix by Segment:** Digital marketing emphasis on Cluster 3, store focus on Cluster 0; balanced omnichannel for Cluster 2.
4. **Develop Targeted Product Lines:** Cluster 0 (families) requires value/bulk offerings; Cluster 1 demands premium products; Cluster 3 benefits from family-oriented bundles.

5. **Periodic Re-segmentation:** Implement quarterly or semi-annual re-clustering using updated customer data to track segment evolution and capture emerging patterns.

7.3 Final Remarks

Customer segmentation through modern unsupervised learning provides quantitative foundation for strategic marketing decisions, replacing intuition-based approaches with data-driven personas. The identified four segments offer immediate applicability for customer lifetime value optimization, churn prediction, and personalized recommendation systems. Continued refinement through predictive modeling and causal analysis will enhance segmentation value in competitive marketing environments.

A Data Description

A.1 Feature List

The dataset comprises 2,240 customer records with 29 features:

Listing 1: Dataset features

```
1 Demographics:
2 - Year_Birth: Birth year of customer
3 - Income: Annual household income
4 - Kidhome: Number of children at home
5 - Teenhome: Number of teenagers at home
6
7 Spending (by product category):
8 - MntWines: Amount spent on wine products
9 - MntMeatProducts: Amount spent on meat products
10 - MntFruits: Amount spent on fruit products
11 - MntFishProducts: Amount spent on fish products
12 - MntSweetProducts: Amount spent on candy
13 - MntGoldProds: Amount spent on gold products
14
15 Purchasing Behavior:
16 - NumDealsPurchases: Number of purchases at discount
17 - NumWebPurchases: Purchases made via website
18 - NumCatalogPurchases: Purchases via catalog
19 - NumStorePurchases: Purchases in physical store
20 - NumWebVisitsMonth: Website visits per month
21 - Recency: Days since last purchase
22
23 Campaign Response:
24 - AcceptedCmp1-5: Binary response to 5 campaigns
25 - Response: Response to most recent campaign
26 - Complain: Number of complaints filed
27
28 Administrative:
29 - Dt_Customer: Date of customer registration
30 - Z_CostContact: Cost of contacting customer
31 - Z_Revenue: Revenue from customer
```

B Implementation Details

B.1 Software Stack

- Python 3.10
- scikit-learn 1.3.0 (clustering, dimensionality reduction)
- pandas 2.0.0 (data manipulation)
- numpy 1.24.0 (numerical computing)
- matplotlib 3.7.0 (visualization)

B.2 Reproducibility

All analyses employ fixed random seeds (`random_state=42`) enabling deterministic results. Code is version-controlled and documented with inline comments. Raw and processed data files are archived for auditability.

C Supplementary Tables

Table 6: Full cluster profile statistics.

Metric	C0	C1	C2	C3
n	1021	131	509	579
Income (mean)	34.7k	79.9k	74.8k	56.9k
Age (mean)	41.9	43.6	45.8	50.9
TotalSpending (mean)	96	1574	1307	669
TotalPurchases (mean)	5.8	19.8	19.6	16.5
Recency (mean, days)	50	49	50	47
CampaignsAccepted (mean)	0.09	2.50	0.26	0.21
WebPct (mean %)	36	28	27	37
Kidhome (mean)	0.81	0.08	0.05	0.23
Teenhome (mean)	0.44	0.24	0.23	0.93